Technische Universität Hamburg

Data Science Foundations

Low-dimensional Control under Embodiment Constraints: From LQR to Reinforcement Learning

Bachelor's Thesis
am Institut Data Science Foundations
Technische Universität Hamburg

vorgelegt von
Alexander Kranz
am
08.09.2026

Erstprüfer:	Prof. Dr. N. Ay
Zweitprüfer	Dr. Manfred Eppe
Betreuung:	Dr. Adwait Datar

Alexander Kranz
geboren am: 09.02.2004
Matrikelnummer: 590156
Studiengang: Data Science

Abstract

Your English abstract here (mandatory if written in English and recommended otherwise).

Zusammenfassung

Hier die deutsche Zusammenfassung einfügen (notwendig).

Contents

1 Introduction **1**

2 Preliminaries **3**
2.1 Control Theory Basics 3
2.2 Notation 3

3 Background **5**
3.1 Motivation 5
3.2 Approach 5

4 Robustness of LQR with low-dimensional Input **7**
4.1 Setting 7
4.2 Results 7
4.3 Discussion 7

5 Robustness under artificially inflated Input **9**
5.1 Setting 9
5.2 Results 9
5.3 Discussion 9

6 Perception-based Experiments **11**
6.1 Setting 11
6.2 Results 11
6.3 Discussion 11

7 Conclusion **13**

A Nomenclature **15**

B Additional Proofs **17**

C Complete Simulation Results **19**

Bibliography **21**

List of Figures

List of Tables

Chapter 1

Introduction

Over the recent years, the field of robotics has seen rapid advancements in the capabilities of neural-network-based implementations. An increase in global compute power [**?**], fueled by the so-called "AI race" of LLMs, has allowed researchers around the world to create even more extensive model architectures for more complex use cases [**?**]. Transferring these structuresto physical bodies requires more space and on-device compute, often the limiting factors of embodiment. To bridge this gap between powerful models and limited resources, the approach of cheap design has become more common. As mentioned by N. Ay [**?**], the core idea is to maintain an "embodied universal approximator", while reducing the overall complexity of the model, which can be defined based on the specific case. The most straightforward approach is inducing the underlying physical dynamics of the embodiment into the architecture, which is often of lower dimensionality than the actual sensors and actuators. Recent papers focusing on algorithms for the LQR [**?**], apply this approach by offering implementations for enforcing low-rank controllers, leading to lower complexity in relation to parameterization. Still, this low-dimensional approach has yet to be extended [**?**] to more complex embodiments, and is missing a crucial part required for implementation: Robustness. This leaves us with the question: *Is it possible to train a low-dimensional controller for an embodied agent, that is performant yet robust?*

As robustness is a wide topic, in our case, we use it when referring to sensor- and actuator noise, as well as parametric uncertainties of the underlying model and external disturbances. When talking about performance, we mean reducing cost as much as possible in relation to an optimal controller. In the end, we want to show in which cases low-dimensional controllers are a viable alternative, and can thus reduce costs for hardware implementations.

To answer our question, we will first explain the core concepts and collect existing approaches for the LQR setting in section 2 and 3. In section 4, we will compare these implementations using the double integrator problem for LQR with respect to general performance, as well as robustness. Building on that, we will

artificially inflate our inputs and outputs via matrix multiplication in section 5 and additionally introduce deep RL baseline models into the comparison. For section 6, we will completely move away from LQR, and extend the setting to a perception-based implementation. Here we will compare different models with varying parameter complexity in regard to robustness, applying the concepts we learned from our earlier experiments. Finally, we will conclude by analyzing whether we indeed achieved our goal of constructing a robust controller, and stating how our approach can be applied to more complex settings.

Chapter 2

Preliminaries

2.1 Control Theory Basics

Information about state space models, LQR, etc.

2.2 Notation

Which letters I will use in what context

Chapter 3

Background

3.1 Motivation

Reviewing content of past papers, state of the art, and what the open problems are.

3.2 Approach

What I will do in the following sections

Chapter 4

Robustness of LQR with low-dimensional Input

4.1 Setting

4.2 Results

4.3 Discussion

Chapter 5

Robustness under artificially inflated Input

5.1 Setting

5.2 Results

5.3 Discussion

Chapter 6

Perception-based Experiments

6.1 Setting

6.2 Results

6.3 Discussion

Chapter 7

Conclusion

You generally cover three things in the Conclusions section, and each of these usually merits a separate subsection:

1. Conclusions
2. Summary of Contributions
3. Future Research

Conclusions are not a rambling summary of the thesis: they are short, concise statements of the inferences that you have made because of your work. It helps to organize these as short numbered paragraphs, ordered from most to least important. All conclusions should be directly related to the research question stated in the introduction. Examples:

1. The problem stated in the introduction has been solved: as shown in Sections ? to ??, an algorithm capable of handling large-scale Zylon problems in reasonable time has been developed.
2. The principal mechanism needed in the improved Zylon algorithm is the Grooty mechanism.
3. Etc.

The Summary of Contributions will be much sought and carefully read by the examiners. Here you list the contributions of new knowledge that your thesis makes. Of course, the thesis itself must substantiate any claims made here. There is often some overlap with the Conclusions, but that's okay. Concise numbered paragraphs are again best. Organize from most to least important. Examples:

1. Developed a much quicker algorithm for large-scale Zylon problems.
2. Demonstrated the first use of the Grooty mechanism for Zylon calculations.
3. Etc.

The Future Research subsection is included so that researchers picking up this work in future have the benefit of the ideas that you generated while you were working on the project. Again, concise numbered paragraphs are usually best.

Appendix A

Nomenclature

What goes in the appendices? Any material which impedes the smooth development of your presentation, but which is important to justify the results of a thesis. Generally it is material that is of too nitty-gritty a level of detail for inclusion in the main body of the thesis, but which should be available for perusal by the examiners to convince them sufficiently. Examples include program listings, immense tables of data, lengthy mathematical proofs or derivations, etc.

Appendix B

Additional Proofs

Appendix C

Complete Simulation Results

Bibliography

Erklärung der Urheberschaft

Ich versichere an Eides statt, dass ich die Bachelor's Thesis im Studiengang Data Science selbstständig verfasst und keine anderen als die angegebenen Hilfsmittel – insbesondere keine im Quellenverzeichnis nicht benannten Internet-Quellen – benutzt habe. Es wurden keine Quellen genutzt, die nicht genannt wurden. Alle Stellen, die wörtlich oder sinngemäß aus Veröffentlichungen entnommen wurden, sind als solche kenntlich gemacht. Ich versichere weiterhin, dass ich die Arbeit vorher nicht in einem anderen Prüfungsverfahren eingereicht habe und die eingereichte schriftliche Fassung der auf dem elektronischen Speichermedium entspricht.

Ort, Datum Unterschrift

Erklärung zur Veröffentlichung

Ich erkläre mein Einverständnis mit der Einstellung dieser Bachelor's Thesis in den Bestand der Bibliothek.

Ort, Datum Unterschrift